

Assignment

Module:	Dept I BAA FS26
Title:	Algorithmic Texture Synthesis: Procedural, Rule-Based and Optimization-Based Methods in Computer Graphics
Background and Problem Statement	In computer graphics, texture mapping is a widely used technique for the realistic rendering of surfaces. Besides the use of pre-existing image textures, algorithmic texture synthesis provides a flexible alternative in which textures are generated automatically. Procedural, rule-based, and optimization-based methods can be applied for this purpose. A central challenge is to generate textures that approximate a given target texture as closely as possible. Since algorithmically generated textures differ structurally from concrete image templates, an exact reproduction is generally not feasible. Instead, the problem can be formulated as an optimization task in which suitable modeling and parameter estimation techniques are used to achieve the best possible approximation. Within the scope of this thesis, it will be investigated to what extent such approaches can be implemented and how well target textures can be approximated using procedural, rule-based, and optimization-based methods.
Objectives and Expected Results	- Research of existing systems for algorithmic texture synthesis and the current state of research - Definition of a suitable use case - Systematic investigation and implementation of: procedural methods, rule-based methods, optimization-based methods for parameter estimation - Development of a parameterizable texture model - Investigation of how textures can be controlled through appropriate parameters and approximated to target textures - If feasible, implementation of an optimization method to minimize the deviation between generated and desired textures using a suitable, defined metric - Comparative evaluation of the approaches with respect to: visual quality, flexibility, reproducibility, parameter controllability, suitability for typical computer graphics applications
Desired Methods and Approach	- Research - Agile development of a prototype - Implementation - Evaluation of the achieved results
Creativity, Methods, Innovation	
Other Remarks:	- Very strong programming skills - Very good knowledge of computer graphics

Project Team

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